组会报告

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2020年3月12日

Motivation

Shuhong Cheng, Jiyong Ma, and Shijun Zhang "Smoke detection and trend prediction method based on Deeplabv3+ and generative adversarial network," Journal of Electronic Imaging 28(3), 033006 (14 May 2019).

火灾保护不仅依赖于早期警报，也依赖于快速有效的火灾救援。在这个过程中，烟雾位置、大小和浓度的变化可以改变实施计划。

本文对烟雾状态进行分析，烟雾浓度在分割区域中通过热图显示。

视频预测用于火灾保护领域，使用 GAN 预测烟雾趋势，来帮助消防员建立合理的救援路线和救援计划。

Method

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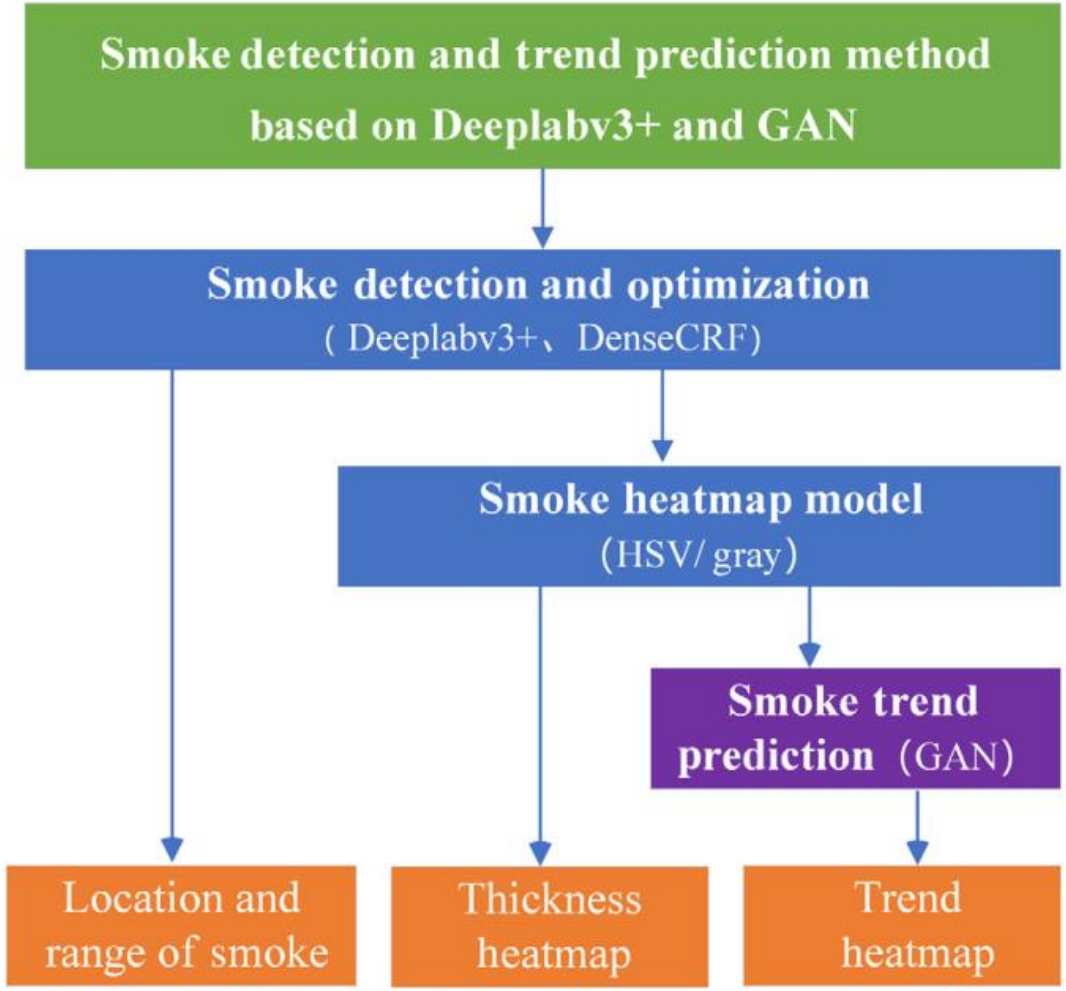

方法流程

Architecture

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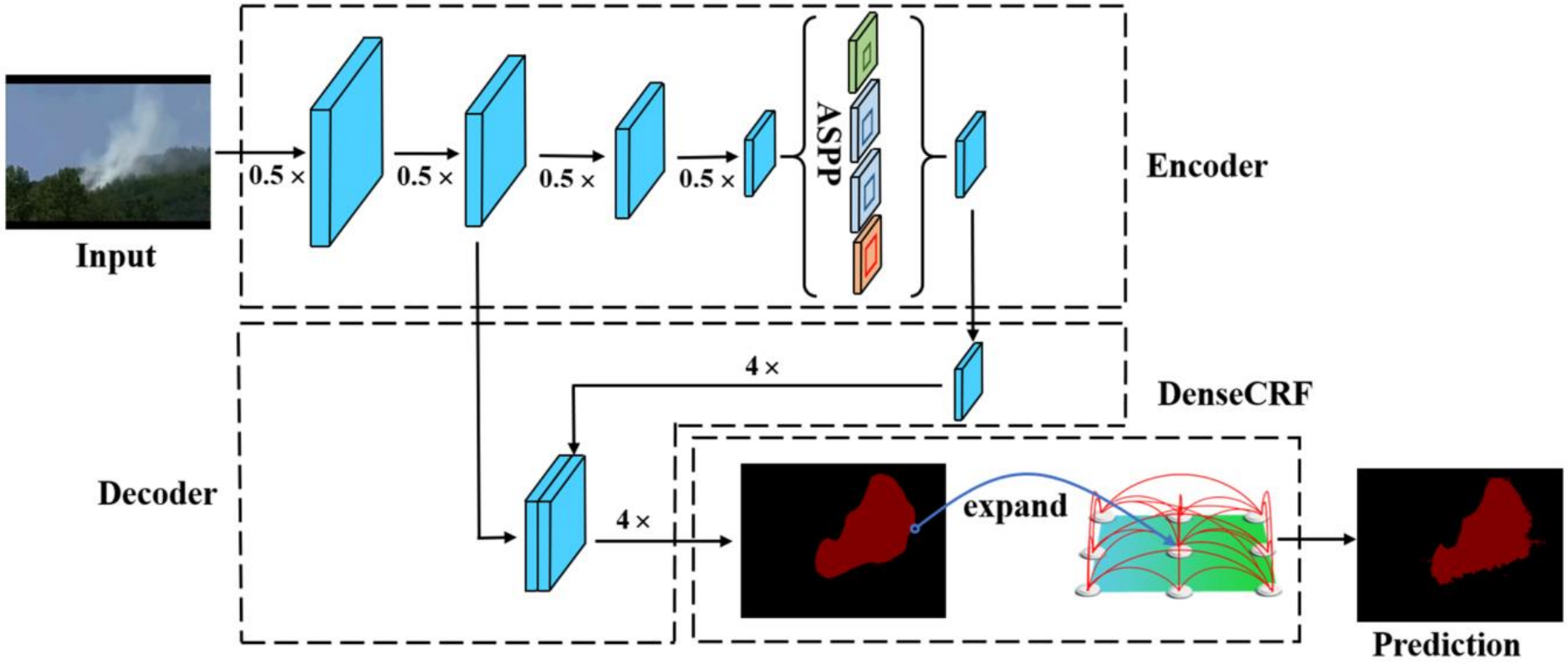

Backbone: DeepLabV3+

ASPP 使用深度可分卷积

DenseCRF 后处理

DenseCRF

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使用 CRF（Conditional Random Field） 来建立任意像素间的关系。CRF 使用烟雾边缘像素作为节点，像素间关系作为边。

CRF 遵循 Gibbs 分布，E 为能量函数： $P(X=x|I)=\frac{1}{Z(I)}\exp[-E(x|I)],$

E 中包含一元项和二元项： $E(x)=\sum_i \psi_u(x_i)+\sum_i \psi_p(x_i,x_j),$

一元势能项表示像素 x_i 的预测结果，二元势能项表示像素 x_i 和其他烟雾边缘像素 x_j 的关系。

$$\psi_p(x_i,x_j)=\mu(x_i,x_j)\sum_{m=1}^{K} w^{(m)}k^{(m)}(f_i,f_j)$$

μ(x_i, x_j) 表示两个像素间的结合力，k^(m)(f_i, f_j) 表示两个像素间的亲密度。

$$k^{(m)}(f_i,f_j)=w^{(1)}exp(-\frac{|p_i-p_j|^2}{2\theta_\alpha^2}-\frac{|I_i-I_j|^2}{2\theta_\beta^2})+w^{(2)}exp(-\frac{|p_i-p_j|^2}{2\theta_\gamma^2})$$

Heatmap

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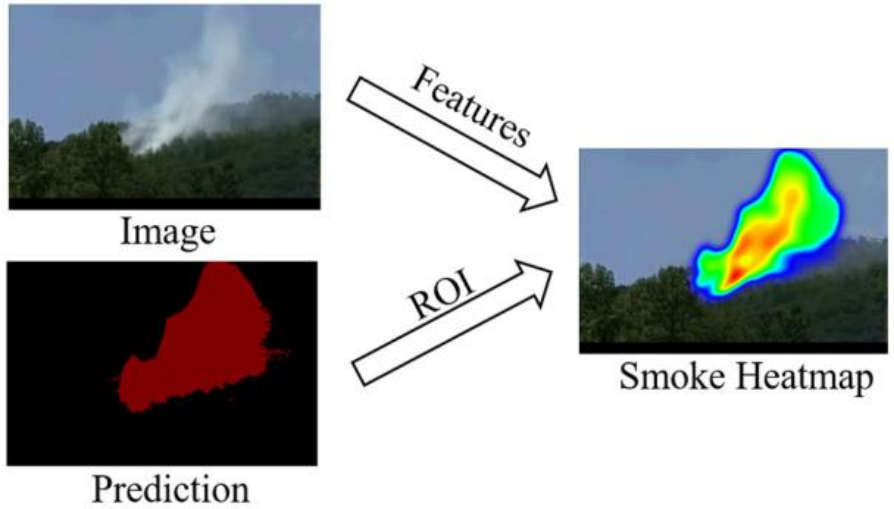

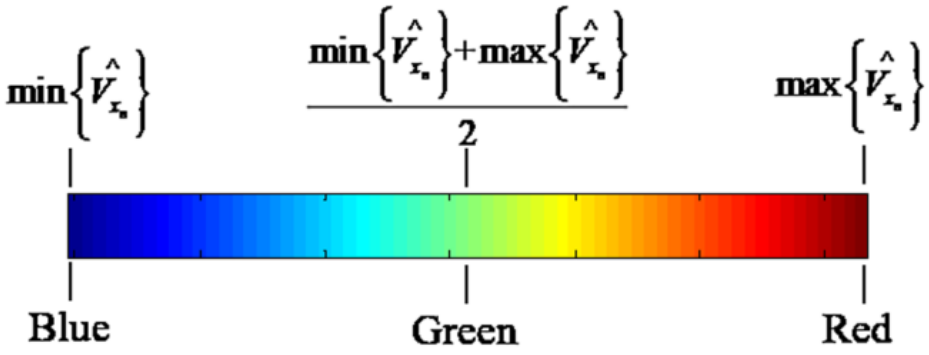

Fig. 5 Principle of rainbow ribbon.

选取 HSV 和灰度特征来表征烟雾的浓度特征。基于 HSV 的烟雾热图原理如下:

1. 烟雾分割方法预测烟雾区域 U，区域包含 n 个像素。
2. 对像素 x_i 设置半径为 10 的缓冲区。
3. 计算像素 x_i 在原始烟雾图像对应位置的 V 值（HSV 的第三个通道值），V=max(R,G,B)。
4. min-max 归一化 。
5. 像素 x_i 根据彩虹色带填入缓冲区。

Trend Prediction

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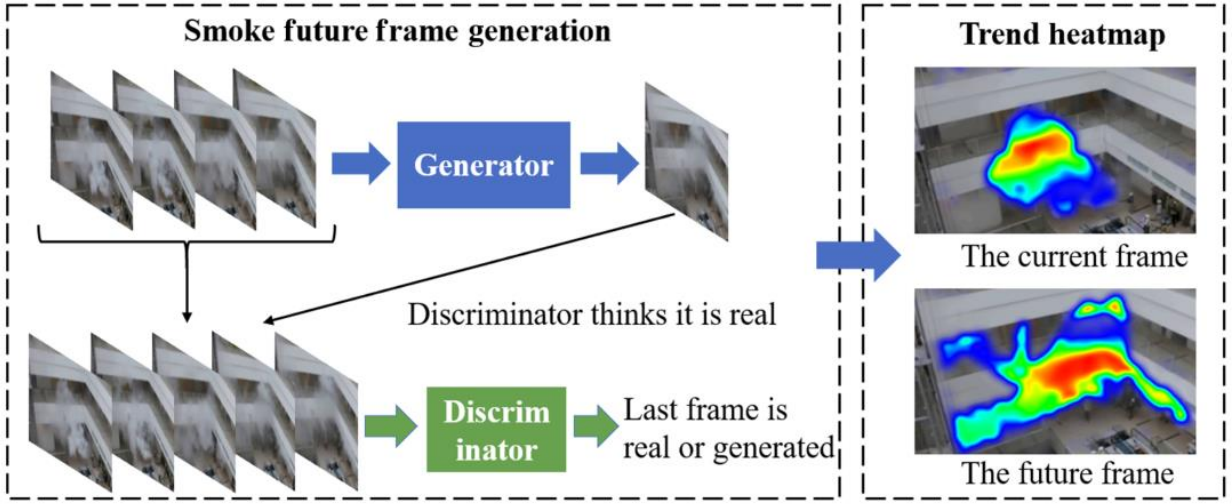

Fig. 6 Smoke trend prediction.

$$G(X) = \{X^{m+1}, X^{m+2}, \ldots, X^{m+n}\}.$$

At the assumption of $m = 4$, $n = 4$

$$\begin{cases} \text{input}: X^1, X^2, X^3, X^4 \rightarrow \text{output}: X^5 \\ \text{input}: X^2, X^3, X^4, X^5 \rightarrow \text{output}: X^6 \\ \text{input}: X^3, X^4, X^5, X^6 \rightarrow \text{output}: X^7 \\ \text{input}: X^4, X^5, X^6, X^7 \rightarrow \text{output}: X^8 \end{cases}.$$

生成器输入是 m 帧的视频序列，X_g 是生成的视频帧。迭代生成 X_g。

Segmentation Experiment

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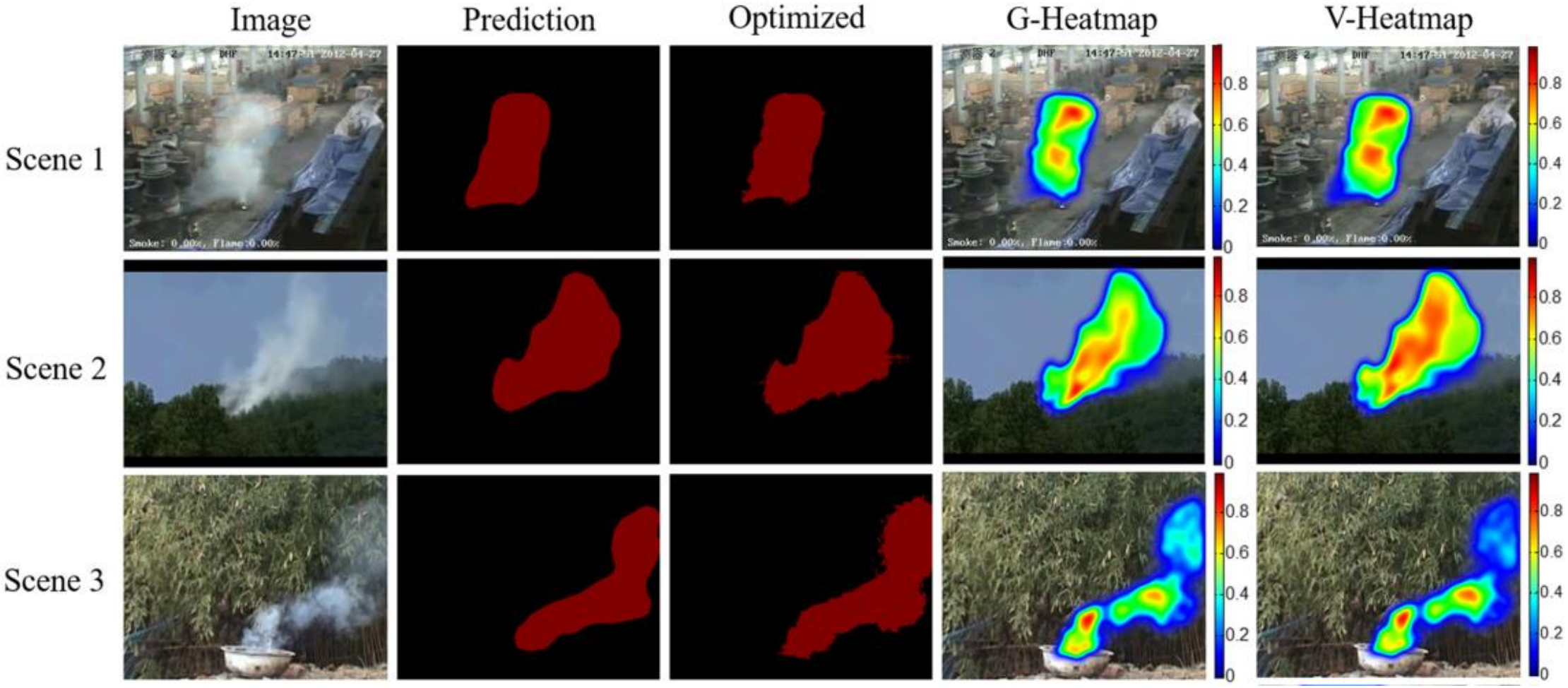

Generation Experiment

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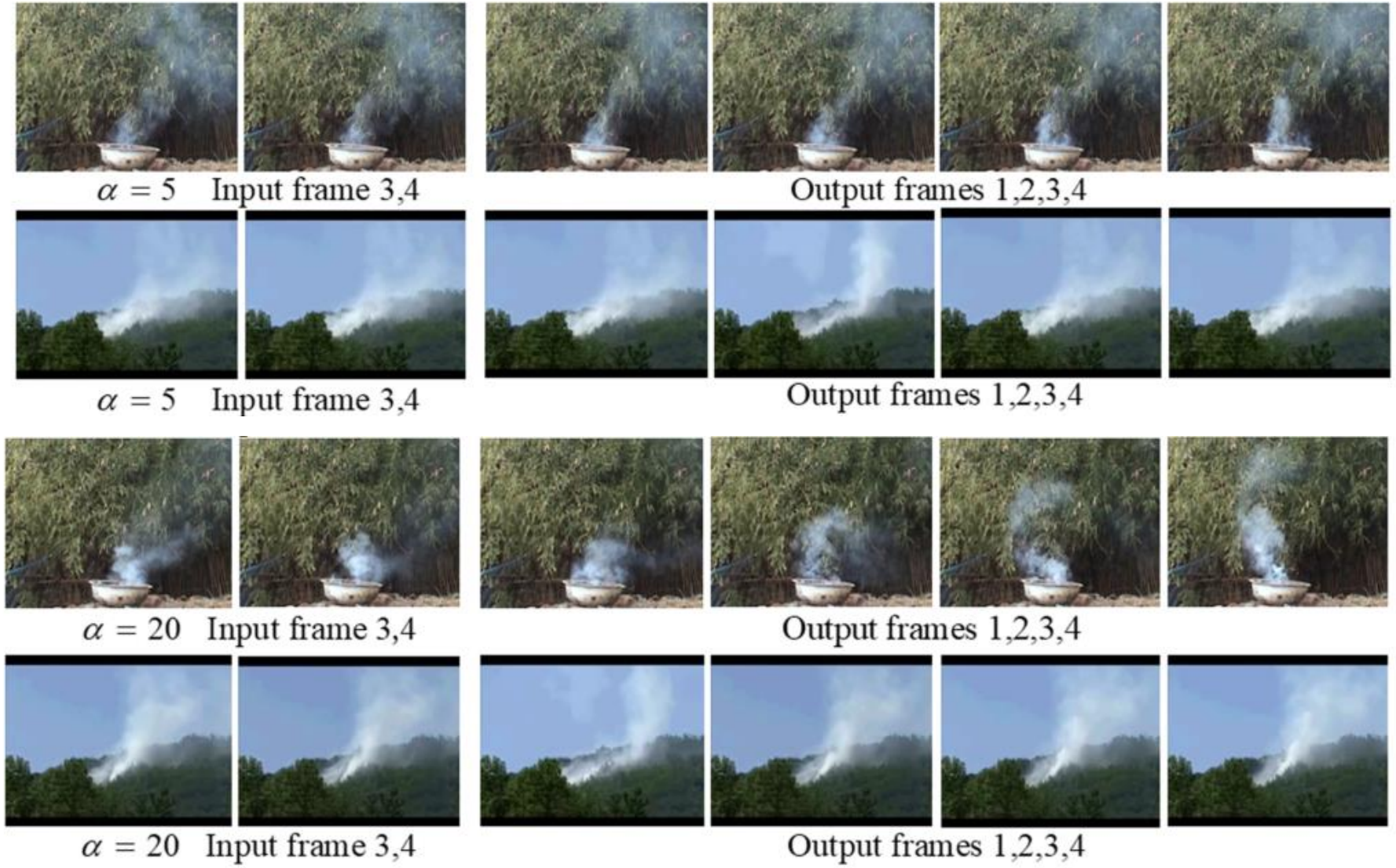

Metric

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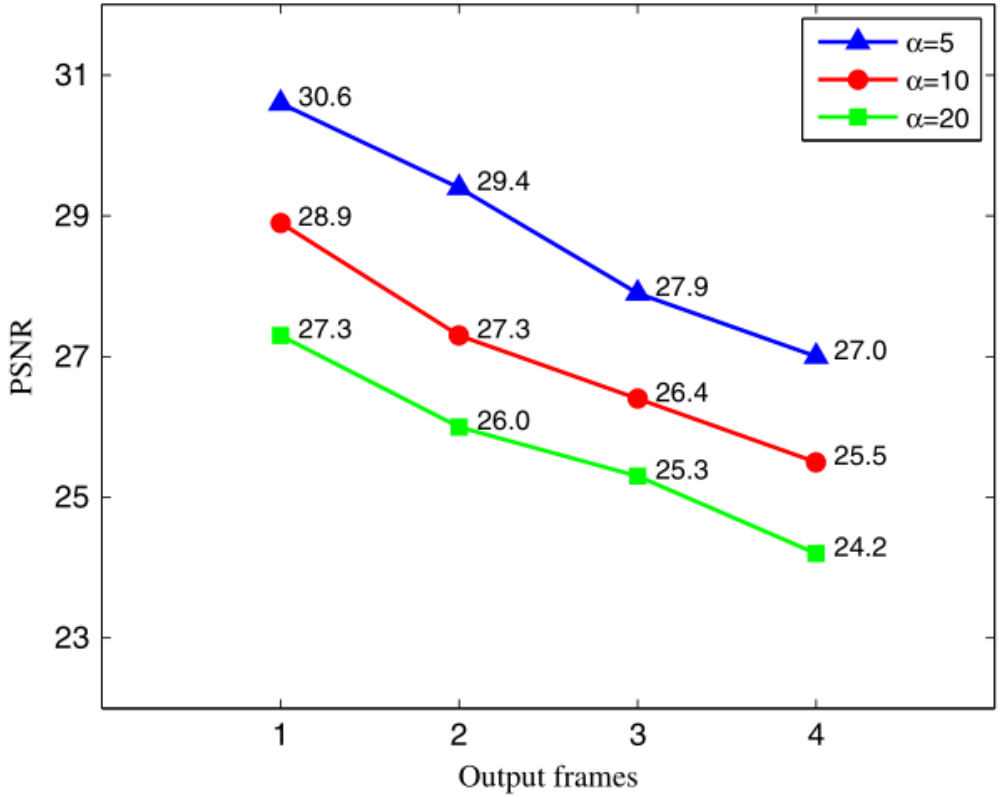
使用峰值信噪比来度量生成的烟雾质量： $\mathrm{PSNR}(Y,\hat{Y}) = 10\log_{10}\frac{\max_{\hat{Y}}^2}{\frac{1}{N}\sum_{i=0}^{N}(Y_i-\hat{Y}_i)^2}$,



a=5 或 10 时，质量好，但是帧间运动状态变化不大。

a=20 时质量差，但是帧间运动状态变化明显，可以有效表达短期烟雾趋势。